Harmonic Distortion Notes: The Two-Parameter Sine Model

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Abstract

We study the simplest two-parameter oscillation and ask how its amplitude and frequency controls interact when a signal is sampled at fixed intervals. Throughout the note the amplitude is written a and the angular frequency is written b; the independent variable is time, written x.

1 The model

The signal of interest is the pure sine with modulated amplitude:

$$y = a \sin(bx). \tag{1}$$

Here a controls the peak value of the signal and b controls how fast the oscillation repeats. Doubling a stretches the curve vertically; doubling b compresses it horizontally by one half. When a is negative the curve flips.

2 Envelope of a damped variant

For comparison, the damped variant decays exponentially:

$$u(t) = e^{-t} \cos(\omega t). \tag{2}$$

Both formulas are studied numerically in the accompanying worksheet; the sweep below fixes the sampling grid and varies the two parameters of (1).

3 A sampled trace

A short sample of the model with $a = 1.5$ and $b = 2$ is plotted in Figure 1.

*Preliminary draft; worked example.

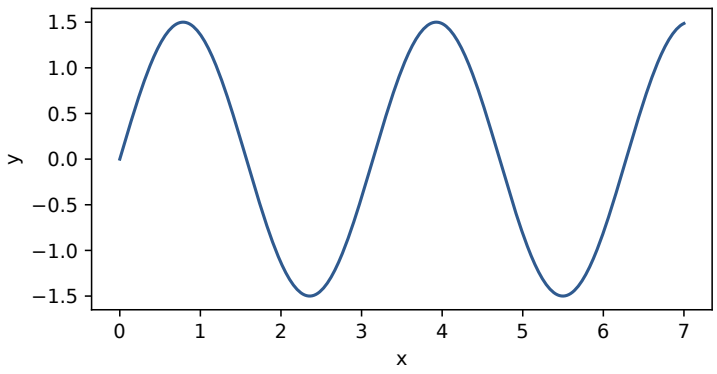

Figure 1: One period-sampled trace of (1) at $a = 1.5$, $b = 2$.